

Artificial Intelligence (AI) Project Proposal



Spring 2026

SECTION BCS-4K

PROJECT NAME: VeriFact: Autonomous Research Agent for Information Verification

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1. Project Overview

- **Project Topic:** A multi-stage autonomous AI agent built using n8n and the Gemini 1.5 Flash API. Unlike standard chatbots that rely on internal training data (which can be outdated or hallucinated), VeriFact actively browses the web to verify claims and cross-references multiple sources before providing a "grounded" response.
- **Objective:** The main goal is to develop an agentic workflow that reduces "AI slop" by automating deep research. The system will take a user query, decompose it into search intents, retrieve real-time data via search APIs, and synthesize a verified report with clickable citations.

2. System Description

- **Original Problem Background:** In the current era of "AI slop," LLMs frequently generate confident but incorrect information (hallucinations). Furthermore, users often suffer from "information overload" when trying to manually verify news or technical claims across dozens of browser tabs.
- **Innovations Introduced:** * **Agentic Decomposition:** The system doesn't just "answer"; it first thinks about what it needs to *know* before answering.
 - **Live Web Grounding:** Integration with DuckDuckGo/Google Search APIs to fetch the latest data.
 - **Source Citation:** Every claim made by the AI is mapped back to a specific URL retrieved during the search phase.
 - **Efficiency:** It automates a 30-minute research task into a 30-second automated execution.

3. AI Approach & Methodology

- **Search Space & Algorithms:** The agent uses a **Decision Tree** logic within n8n to determine if a search result is relevant. It implements a basic **Breadth-First Search (BFS)** across the first page of search results to gather a diverse set of viewpoints.
- **AI APIs:** * **Gemini 1.5 Flash (via Google AI Studio):** Chosen for its massive context window and fast reasoning (Free Tier).
 - **DuckDuckGo Search Node:** Used for real-time web retrieval without requiring an API key (Free/Open).
- **Automation Tools:** **n8n (Desktop/Self-hosted):** Used to orchestrate the flow between the user, the LLM, and the search tool.
- **Other Techniques:** **Prompt Engineering:** Utilizing "Chain-of-Thought" prompting to force the LLM to verify its own logic before finalizing the report.

4. System Rules & Workflow

- **System Workflow:**
User Input → LLM (Query Generation) → n8n (API Search Calls) → LLM (Information

Synthesis & Fact-Check) → Final Response (Report with Citations)

- **Output / Success Conditions:** A successful execution results in a report that:
 1. Directly answers the user's claim.
 2. Labels the information as "Verified," "Inconclusive," or "False."
 3. Lists at least three distinct web sources as references.
- **Process Sequence:**
 1. **Trigger:** n8n receives a question via a Chat Trigger or Webhook.
 2. **Plan:** Gemini generates 3 specific search queries to cover different angles of the question.
 3. **Search:** n8n iterates through these queries using the DuckDuckGo node.
 4. **Extract:** A "Code Node" parses the JSON results to extract snippets and URLs.
 5. **Synthesize:** Gemini receives the original question + the 10+ search snippets and writes the final verified answer.

5. Libraries and Tools

- **AI Models:** Google Gemini 1.5 Flash API.
- **Automation:** n8n Desktop.
- **APIs:** DuckDuckGo Search (Built-in n8n node).
- **Python/JS:** Small JavaScript snippets within n8n "Code Nodes" for data formatting.